

specifications. Dhrystone (DMIP) benchmark is one of the measurement methods for CPU performance. The code tries to twist and test various parts of the CPU core. A higher value of DMIP/MHz means a better and efficient core. This provides a simple means for most developers to choose a core for their project.

Binu Raj S., senior architect-hardware design team, Dexcel Electronics Designs, shares, "A lot of analysis is done before freezing on to a particular MCU that best fits the requirements in the initial phase of the project. Features, integrated peripherals, performance, cost and power are some of the key decision factors for selection of a particular MCU." Tools and support, code reuse and design portability are the other considerations.

LIM, Wee Seng, senior manager, Renesas Electronics, says, "Advanced developers will peek into the CPU core architecture to seek an answer to their higher level of problems. This can be the need of floating-point unit (FPU), DSP-like instruction, multiply-and-divide instructions for many mathematical calculations, fast interrupt latency for fast event response, more number of 32-bit registers for 32-bit data manipulation and so on."

With optimal performance, energy efficiency becomes a key factor in em-

What embedded designers have to say

"Online support for an MCU is quite important. If standard initialisation codes and interfacing codes are available, it reduces the effort for the designer"

—Praval Jain, VP-networks, Virtual Wire Technologies

"Indian embedded society is one where migration to newly available MCUs is required to apart the cost and performance benefits"

—Upendra Patel, CTO, elfnchips

"Apart from price and support, nowadays we are very careful about the lead time"

—Binu Raj S., senior architect-hardware design, Dexcel Electronics Design

"It's up to the designer's knowledge and imagination as to how he makes optimal use of available resources"

—K.C. Bhasin, Ex-EFY Technical Lab

"My choice of microcontroller is based on the availability of tools and samples as today all the companies are able to offer low-power, low-price parts"

—Vinay Chaddha, CEO, GVC Systems

bedded applications. Sadasivan adds, "In embedded world, it is not just about pushing the MHz speed as in the case of processors. The emphasis is on energy and computing efficiencies." "Imagine you have a device with processor speed of 200 MHz and a flash memory, but the flash memory has a limitation up to 40-50 MHz. So increasing only the processor speeds will not make any difference to your embedded design," he explains.

According to Wee Seng, "Most developers write their programs in 'C' language, and not Assembly language. They are more concerned about how well the core can perform with an efficient compiler, and how easy to use

is the IDE and debugger."

Open source tools, which can be downloaded for free, also drive the MCU selection. Praval Jain, vice president-networks at Virtual Wire Technologies, cites, "If there are free tools available, and their performance is quite similar to paid tools, such MCUs are preferred." You can get a variety of free tools for standard cores and the support community is also huge.

According to Eieland, "There are a number of open source tools available for proprietary cores as well. In fact, you also have open source hardware in the form of Arduino platform based on an MCU which is a proprietary core."

Is there a need for standard core?

A microcontroller architecture develops a huge world of reference designs, application notes, software tools, hardware tools and coding language around it. "The success or failure, and the time to market of a product depend heavily on the strategy of embedded software development," feels Amit Bhojraj, product marketing manager for business line-microcontrollers, NXP Semiconductors. "Though there is still a lot of work involved in moving a design from one MCU vendor to another that offers the same core architecture, it is still considerably easier than to move between proprietary core based microcontrollers," he adds.

8051 and PIC rule in 8-bit

Invention-wise, the forefather of the microcontroller world is 8051. Then came in PIC and Renesas and the most recent is AVR. 8-bit is dominated by 8051 because of its long existence, improvements with time, established tools and familiarity in student community. Though IP provider Intel has discontinued MCS 51, other vendors still offer 8051 flavours of MCUs.

"The launch of Microchip's PIC was a game changer in 8-bit as it offered a very small-size and low-cost MCU. Flash memory based PICs range from tiny 6-pin SOT23 package to 100-pin TQFP package targeted at small-size embedded products," shares Upendra Patel, CTO, elfnchips.

Although PIC and AVR have captured the market, 8051 is still in use because of its simplicity, familiarity and low cost. 8051 leverages a huge supporting community as it is manufactured by around 50 different vendors, whereas PIC and AVR are manufactured by their parent companies Microchip and Atmel, respectively.

Market research firm Gartner ranked Microchip as No. 1 in 8-bit microcontrollers and listed Atmel as the fastest growing 8-bit microcontroller vendor in the world in 2008 with its AVR and Otouch capacitive-touch technology. From applications' perspective, iSuppli reports that one of the key trends for 8-bit market is integration of specific peripherals for dedicated tasks—like the LCD controller, USB ports, touch controllers and extended memories.